

Warehouse disruption repair: POCL + plan modification

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Sources, the test suite and every generated artefact (animated GIFs of both case studies, the repair cards and this report) live at:

https://github.com/aaditya-biswas/AS_Assignment

1. Problem statement and assumptions

A fleet of robots serves pickup/delivery tasks on a shelf grid. Disruptions -- exogenous blockages and endogenous accidents/emergencies -- invalidate the plan; the repair must be local, fast and safety-preserving.

Assumptions: unit cells and 4-neighbour moves, one robot per cell per tick, time-indexed vertex and edge reservations, negotiation only within comm_radius, a bounded repair budget, and no global re-planning inside a repair.

2. Three-layer architecture

Layer	Module	Responsibility (POCL mapping)
L1 partial order	pocl.py	STRIPS operators, threats, POP search
L2 grounding	planner.py / peg.py	space-time A*, PEG retiming
L3 repair	modify.py / negotiation.py	the R0-R4 ladder + protocol

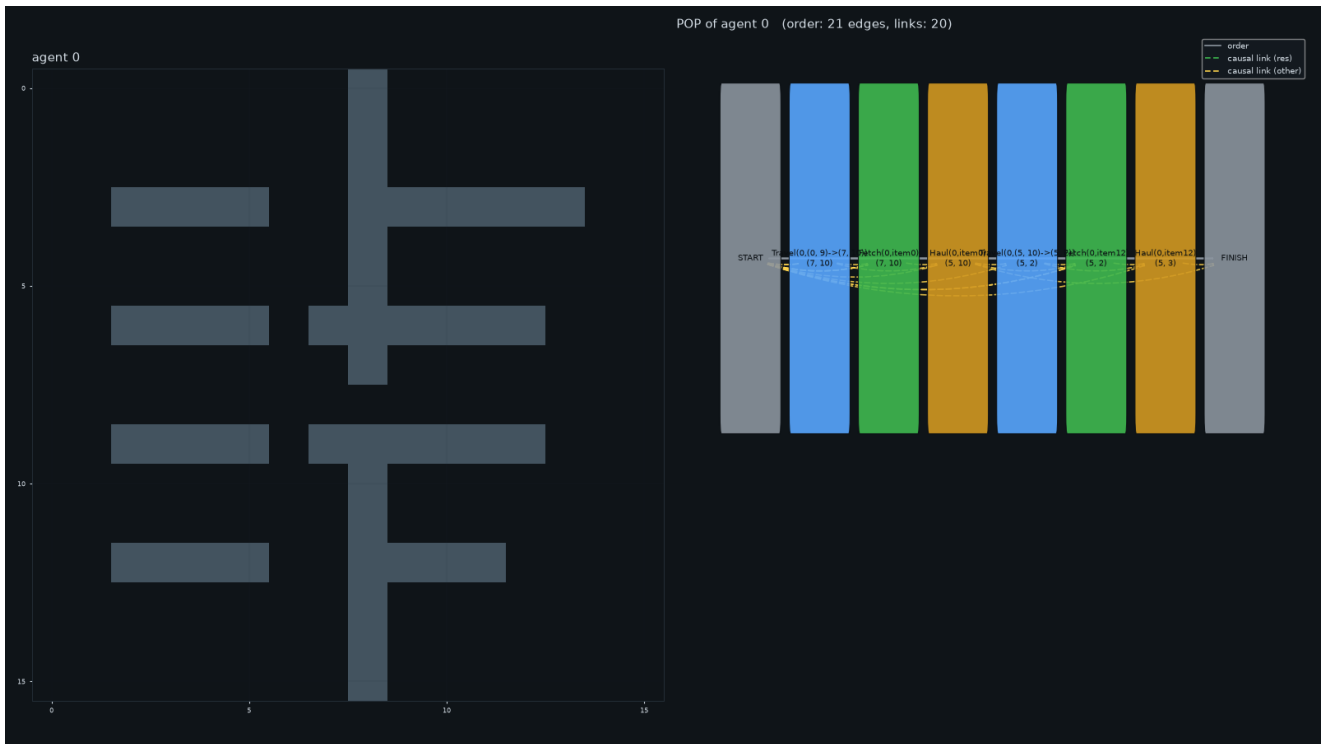
3. Case study A: congested warehouse (aisle closed)

One cross-aisle is closed for maintenance, so the fleet funnels through a single door, and the shift starts with closed lanes, robots down and one emergency diversion. That is the shift `run\_demo.py` animates, so the panels show live counters (altered agents, protocol messages, disruptions, blocked cells, waiting pairs) rather than the five zeros of an all-clear run.

case	tasks	makespan	safe	repairs	altered (sum)	messages	rungs	FAILED_INIT
negotiate, 16x16 grid (aisle closed), N=12	24/24	128	yes	8	32	17	{"R1": 2, "R2": 6}	0

metrics-panel counter	peak in this run
altered agents	12
protocol messages	17
disruptions	9
blocked cells (peak)	3
waiting pairs (peak)	3

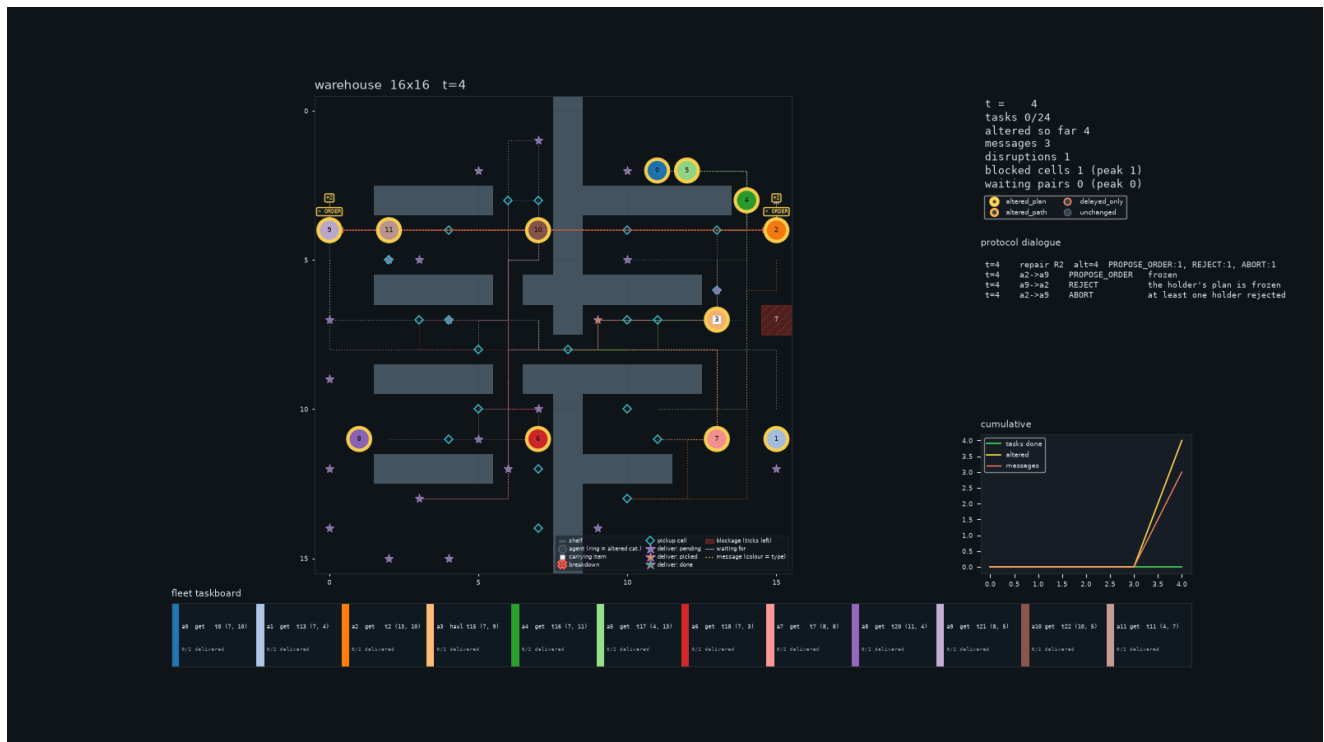
None of the five metrics-panel counters is zero in this run: 12 altered agents, 17 protocol messages, 9 disruptions, up to 3 closed cells at once and up to 3 queueing pairs at once. The animated GIF of the same shift is `outputs/negotiate/animation.gif`.



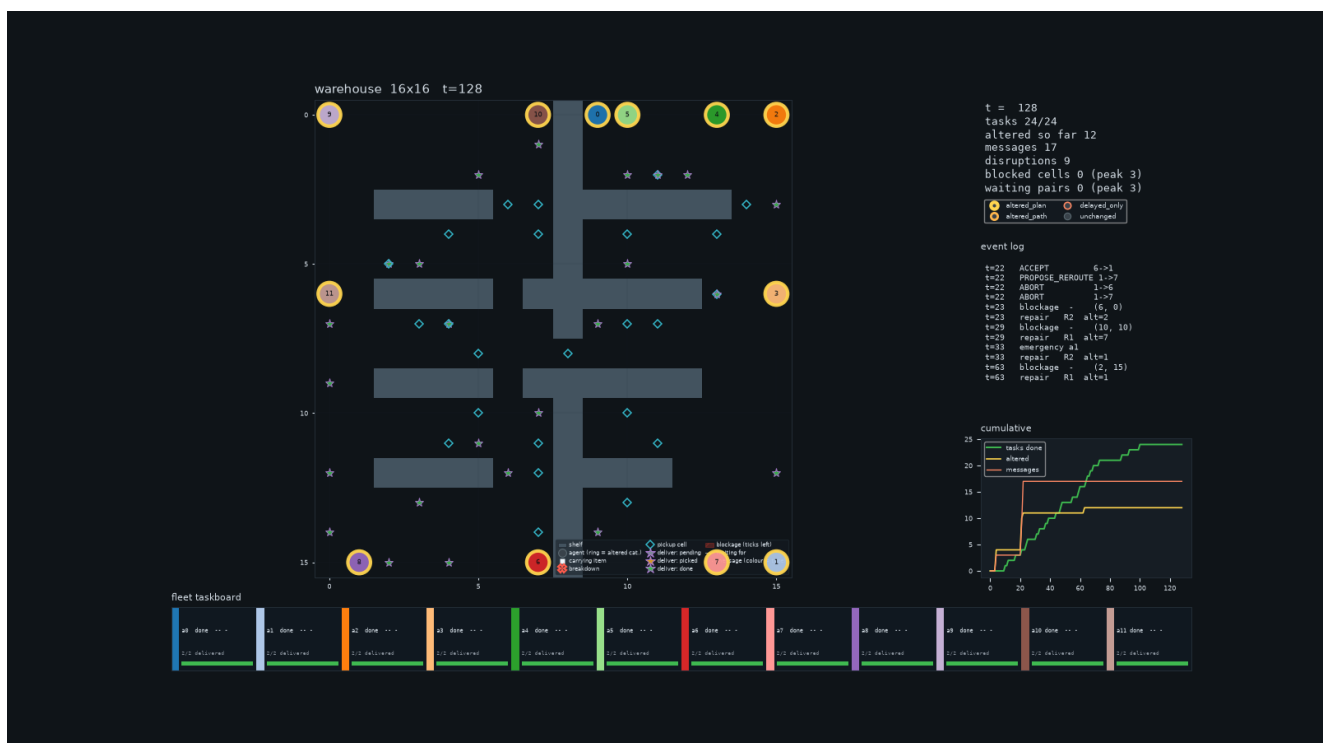
A POP of an altered agent (L1).



Per-agent Gantt of the repaired schedule.



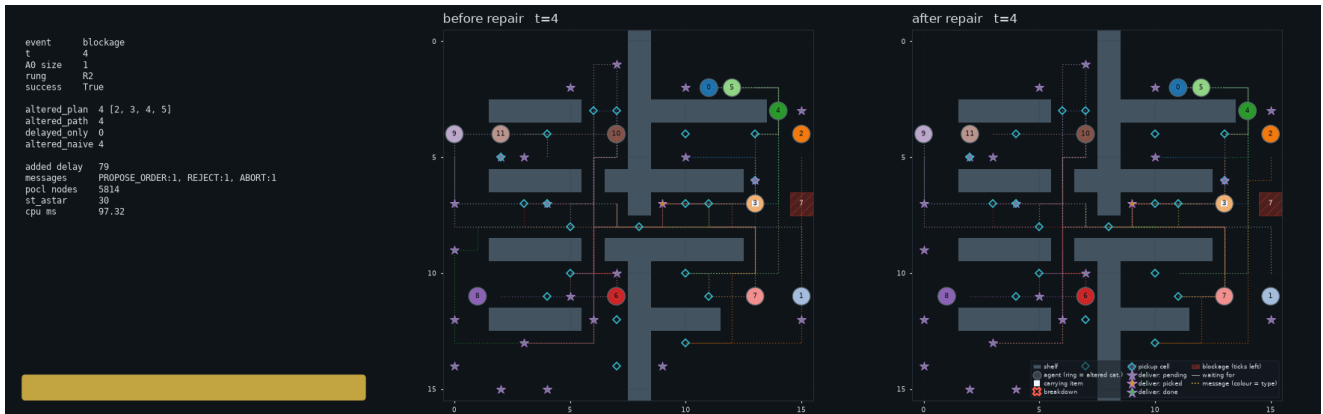
Metrics panel with the protocol dialogue.



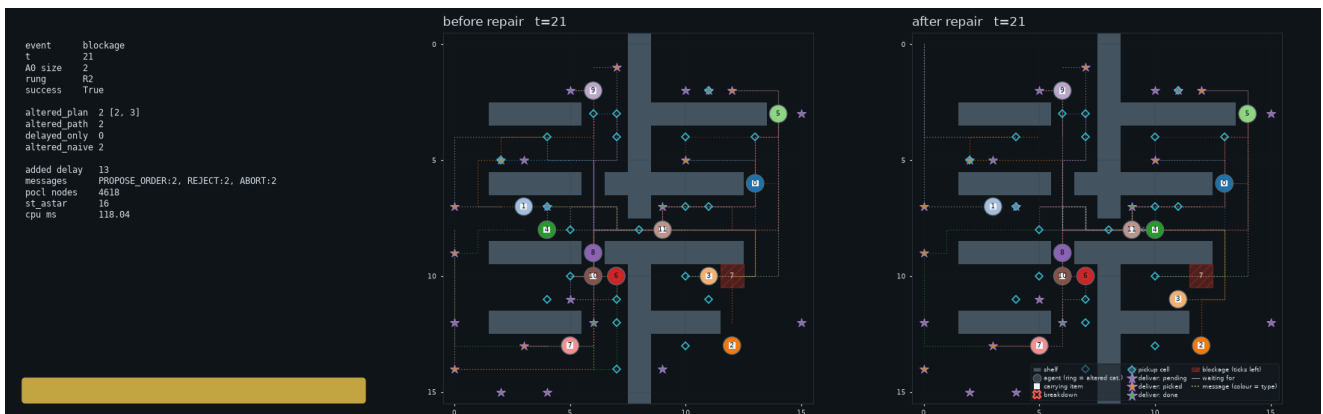
Final state of the fleet.

3.1 Repair cards

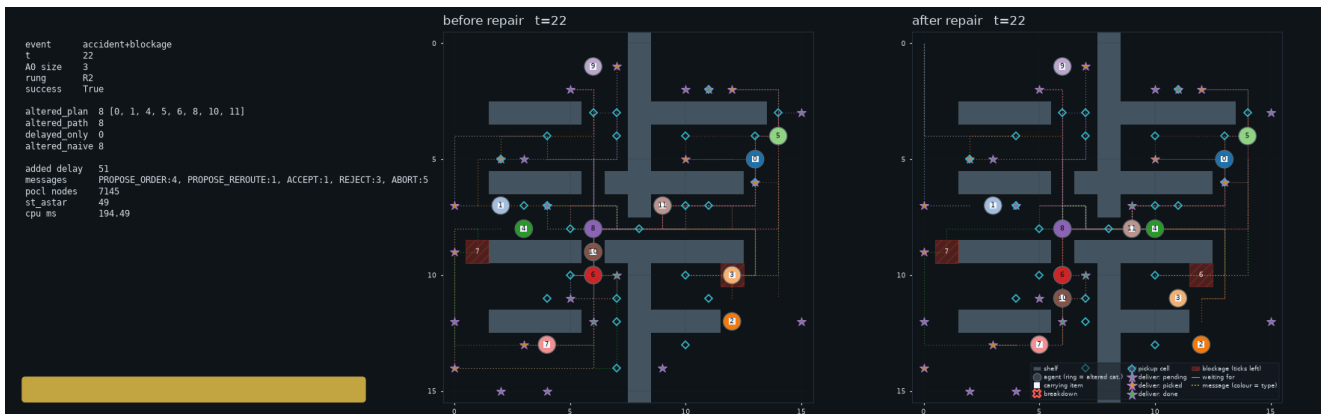
4 repair card(s): the RepairRecord (rung, altered sets, message counts, budgets) beside the routes before and after.



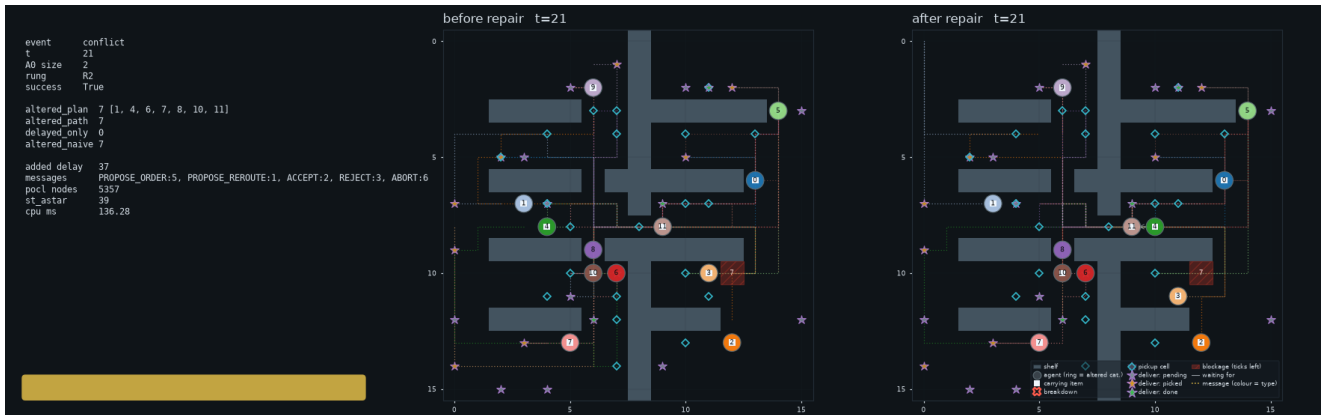
repair card 00 t4



repair card 01 t21



repair card 02 t22



repair card 03 t21

4. Altered agents per disruption (SPEC 9.5)

definition	mean	median	max
altered_plan	4	3	8
altered_path	4.125	3	8
delayed_only	0	0	0
altered_naive	4.125	3	8

t	event	rung	altered_pla n	altered_pat h	delayed_onl y	altered_nai ve	added_dela y	messages (cumulative)	cpu_ms	success
4	blockage	R2	4	4	0	4	79	3	97.3	yes
21	blockage	R2	2	2	0	2	13	6	118	yes
22	accident+bl ockage	R2	8	8	0	8	51	14	194.5	yes
21	conflict	R2	7	7	0	7	37	17	136.3	yes
23	blockage	R2	2	2	0	2	10	17	44.3	yes
29	blockage	R1	7	8	0	8	1	17	86.5	yes
33	emergency	R2	1	1	0	1	24	17	41.9	yes
63	blockage	R1	1	1	0	1	2	17	8.5	yes

5. Case study B: the corridor choke point

A single-door corridor forces a clash, so the ladder finds no hard candidate and the initiator negotiates a grant (rung R2): PROPOSE_REROUTE -> ACCEPT -> COMMIT. Case A now speaks too -- the closed aisle funnels the fleet through one door, so holders clash and the protocol answers -- but the corridor remains the minimal instance where a repair *cannot* be a silent local detour.

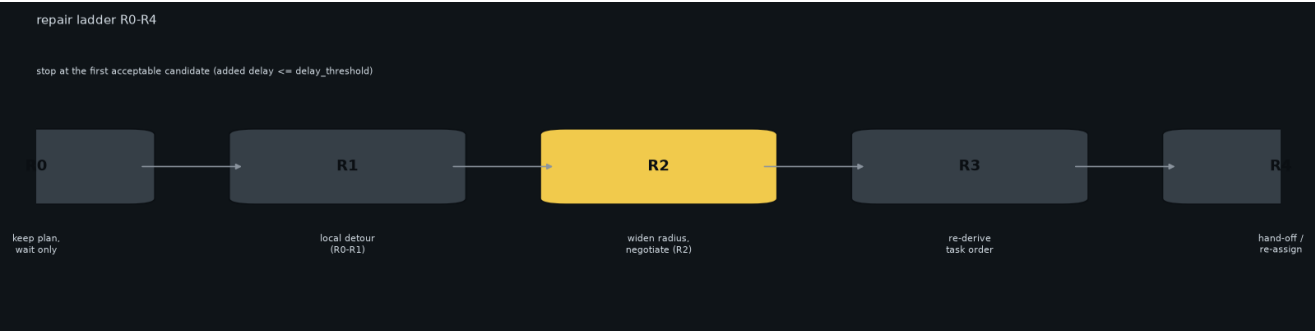
case	tasks	makespan	safe	repairs	altered (sum)	messages	rungs	FAILED_INIT
choke corridor, N=4	3/3	19	yes	1	2	3	{"R2": 1}	0

metrics-panel counter	peak in this run
altered agents	2
protocol messages	3

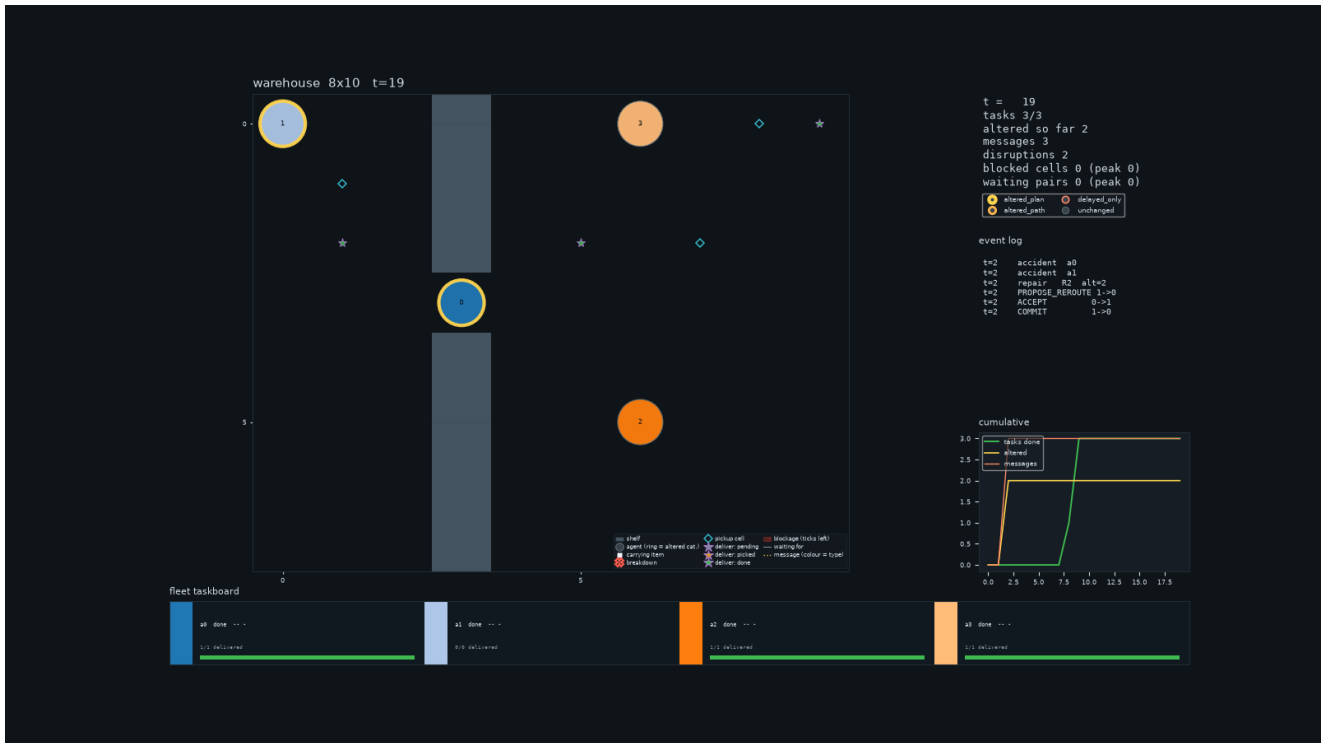
metrics-panel counter	peak in this run
disruptions	2
blocked cells (peak)	0
waiting pairs (peak)	0



The message sequence diagram of the pass.



The R0-R4 ladder, with the used rungs lit.



The corridor at the final tick.

6. Experiments

No sweep data found; run ``python experiments.py --sweep A --seeds 10 --plots`` to fill this section.

7. Discussion

Partial orders plus negotiation keep the altered set small: negotiate alters far fewer agents than the global baseline at a comparable makespan, and the corridor shows the protocol only speaks when a holder really parks in the way. Remaining risks: prioritized-planning incompleteness, deadlock under a fully blocked choke, and the `failure_wait` fallback (rung R4) when no candidate exists.

8. How to run

Everything is reproducible from the published repository:

```
git clone https://github.com/aaditya-biswas/AS_Assignment
cd AS_Assignment && pip install -r requirements.txt
python run_demo.py --compare --viz --outdir outputs
python run_demo.py --choke --compare --viz --outdir outputs/choke
python experiments.py --sweep grid --seeds 10 --plots
python make_report.py --data outputs --zip
```

The animated GIFs of the shifted warehouse are ``outputs/negotiate/animation.gif`` and ``outputs/bfs/animation.gif``; the corridor showcase is ``outputs/choke/negotiate/animation.gif``. Pass ``--open-warehouse`` to run the same shift on the open floor instead of the closed aisle.

9. References

UCPOP (1992); Weld (1994); Kambhampati & Hendler (1992); Fox et al. (2006); Silver (2005); Hoenig et al. (MAPF-POST, 2016).